



University of Engineering and Management
Institute of Engineering & Management, Salt Lake Campus
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Syllabus for B.Tech Admission Batch 2023

Subject Name: Programming using Python Lab

Subject Code: OECEEE481B

Assignments:

Part A

1. Write a program to calculate the square and cube of a number using * operator.
2. Write a program to calculate simple interest (SI). Read the principle, rate of interest and number of years from the user.
3. Write a program to read the weight of an object in grams and display its weight in kilograms and grams, respectively
4. Write a program to reverse a four-digit number using % and // operators.
5. Write a program to read two numbers from the user. Display the result using bitwise | operator on the numbers.
6. Write a program to read two numbers from the user. Display the result using bitwise '&', '~' and '^' operation on the numbers.
7. Write a program to traverse all the elements of a string.
8. Write a program using compound assignment operators to calculate the area of a rectangle.
9. Write a program which prompts a user to enter the radius of a circle. If the radius is greater than zero then calculate and print the area and circumference of the circle.

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10.: Write a program to find out the grade as per the percentage given in following the table.

Percentage	Message
per >= 90	Distinction
per >= 80 && per < 90	First Class
per >= 70 && per < 80	Second Class
per >= 60 && per < 70	First Class
per < 60	Fail

11. Write a program to calculate the salary of a medical representative considering the sales bonus and incentives offered to him are based on the total sales. If the sales exceed or equal to 1,00,000 follow the particulars of Column 1, else follow Column 2.

Column 1	Column 2
Basic = ₹4000	Basic = ₹4000
HRA = 20% of Basic	HRA = 10% of Basic
DA = 110 % of Basic	DA = 110 % of Basic
Conveyance = ₹500	Conveyance = ₹500
Incentive = 10% of Sales	Incentive = 4% of Sales
Bonus = ₹1000	Bonus = ₹500

12. Display capital letters from A to Z.

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13. Use for loop to print numbers from 1 to 10 in the reverse order.
14. Write a program to print squares of the first five numbers.
15. Write a program to print the sum of squares of the first five numbers.
16. Write a program to print even numbers from 0 to 10 and find their sum
17. Write a program to calculate the sum of numbers from 1 to 20 which are not divisible 2, 3 or 5.
18. Write a program to add the sum of digits from 1 to 25, 50 to 76 and 90 to 101 using user created function.
19. Write a program to find the maximum of two numbers.
20. Find the S.I using by creating function based on positional argument .The value of to be provided at the run time
21. Write a function calc Distance (x1, y1, x2, y2) to calculate the distance between two points represented by Point1(x1, y1) and Point2 (x2, y2).
The formula for calculating distance is:
$$\text{Distance} = \sqrt{(x2 - x1)^2 + (y2 - y1)^2}$$
22. Find the factorial of any user given number
23. Create a tuple whose elements are APPLE. Find the length of tuple,the elements that has maximum and minimum value and also print the index position of A and P
24. Write a program to pass a list to a function. Calculate the total number of positive and negative numbers from the list and then display the count in terms of the dictionary.
25. Write a Python program to append a new item to the end of the array ('i', [1, 3, 5, 7, 9]). Append 11 at the end of the array.
26. Write a Python program to reverse the order of the items in the array.('i', [1, 3, 5, 3, 7, 1, 9, 3]).
27. Write a Python program to get the number of occurrences of a specified element in an array ('i', [1, 3, 5, 3, 7, 9, 3]).

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28. Write a Python program to insert a newly created item before the second element in an existing array ('i', [1, 3, 5, 7, 9]) Insert new value 4 before 3.
29. Write a Python program to find a pair with the highest product from a given array of integers.
Original array: [1, 2, 3, 4, 7, 0, 8, 4]
Maximum product pair is: (7, 8)
Original array: [0, -1, -2, -4, 5, 0, -6]
Maximum product pair is: (-4, -6)
30. Write a Python program to create an array of six integers. Print all members of the array.
Sample Output:
10
20
30
40
50
60

Assignments:

Part B

1. Hands-on exploration of core Python data types for analysing 10-year climate dataset.
2. Hands-on experiments to implement and apply arithmetic and logical operators in power calculations of thermal power plant using Python.
3. Design Python indexing techniques in image processing in healthcare.

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4. Developing real-time decision-making systems for healthcare solutions using conditional statements, control flow, and nested loops using Python.
5. Design a dictionary to manage student records, mapping roll numbers to scores using Python.
6. Create and analyse features relevant to the design parameters of solar cells using Python.
7. Design mathematical transformations to preprocess and enhance data for anomaly detection in industrial windmill performance using Python.
8. Create insightful visualizations of data distributions: (a) Scatter plots of customer segmentation for marketing, (b) Heatmaps for supply chain optimization and (c) Trend analysis in stock prices using Python.
9. Use of Eigenvalues and Eigenvectors in Principal Component Analysis (PCA) for dimensionality reduction in machine learning using Python.
10. Matrix Rank in detecting redundancy in industrial sensor data using Python.
11. Create 'Training' and 'Testing' using Python by treating algorithms like decision trees and nearest neighbors as black boxes, focusing on their application in the healthcare domain.